

SaxonLLM

Low-Resource Machine Translation for Saxon with Large Language Models

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1. Introduction

This project initially aimed to investigate LLM-based machine translation (MT) involving Saxon, ideally in both translation directions between German and Saxon. The initial assumption was that Saxon would constitute a low-resource setting in which the availability of suitable training data could be a major limitation. An initial search for existing German-Saxon resources therefore focused on corpora, direct translations of books, and relevant web-based material. However, this search revealed essentially no suitable parallel data that could directly support the intended translation task and therefore proved the initial assumption right.

This lack of German-Saxon parallel data shifted the focus of the project from translation model development toward the construction of a German-Saxon parallel dataset. The main contribution of the project is consequently the creation and curation of a dataset, which can serve as a basis for future experiments with LLM-based translation and adaptation.

2. Background & Related Work

2.1 Saxon dialect

The dialect landscape of present-day Saxony has historically been highly diverse, with researchers in the 20th century originally distinguishing 21 different dialects, which were later grouped into five larger dialect groups. [1]

This historical classification, however, no longer corresponds fully to the present-day linguistic situation. The traditional dialects have undergone considerable decline, while regiolects and regional accents have become more prevalent in everyday speech. In formal situations, regional features are increasingly replaced by standard-language forms. Nevertheless, stronger dialectal and regiolectal forms are still found particularly in smaller municipalities in the Vogtland and Erzgebirge, where several speech varieties can still be observed, including among younger speakers. [1]

According to perceptual dialectological studies, speakers in Saxony distinguish **five broader linguistic regions** today: the Leipzig area and its northern surroundings, the Meißenisch region centered around Dresden, Vogtland, Erzgebirge, and Lausitz. Thus, referring to “Saxon” as a single dialect oversimplifies the regional variation within Saxony. [2]

2.2 Low-resource machine translation

Low-resource machine translation is constrained by the **scarcity of parallel data**, which is particularly relevant to dialectal NLP. As Artemova and Plank [3] show, resource scarcity and the lack of standardized orthography are notable challenges when processing German dialects.

Synthetic parallel data provides one approach to mitigating the data bottleneck. Lin et al. [4], for example, address Nigerian Pidgin by collecting an English-Pidgin parallel corpus and combining it with cross-lingual adaptive training and back-translation. Their results show that such adaptation using synthetic data can improve performance in a low-resource setting. de Gibert et al. [5] demonstrate that LLM-generated synthetic data can improve MT performance for low-resource language pairs, including when used to fine-tune pretrained systems. Pei et al. [6] further show for Manchu that in-context MT can be used to generate synthetic parallel data that improves subsequent low-resource MT.

For the subsequent model tuning, **parameter-efficient fine-tuning** (PEFT) is a potential approach to adapting pretrained models effectively in low-resource translation settings. Su et al. [7] systematically evaluate multiple PEFT methods across low-resource translation tasks and find that several approaches can outperform full fine-tuning.

For the present project, these findings motivate the collection of monolingual (Saxon) and parallel (German-Saxon) data as a basis for future LLM-based translation experiments.

3. Dataset & Methodology

The dataset was constructed from several web-based sources. The main sources were the Saxon Wikipedia hosted in the Wikimedia Incubator, the SimplySaxony Instagram account, and the Sachsen-Lese website. In addition, further Saxon-language material was collected through targeted web searches. Since the available sources differ substantially in format and linguistic characteristics, each source was processed separately.

3.1 Saxon Wikipedia

The first source was the *Saxon Wikipedia* in the Wikimedia Incubator, where the archived version available through the Wayback Machine for 8 August 2024 [8] was used as the reference point. All Wikipedia pages that existed in this archived snapshot were collected. Each webpage was manually extracted and represented as a single dataset entry.

The Wikimedia material was only monolingual Saxon data, and the collected pages were not translations of existing Wikipedia entries but instead adapted or newly created entries.

3.2 SimplySaxony Instagram

The only source of German-Saxon **parallel data** was the *SimplySaxony* Instagram account [9]. The account was identified through web searches and subsequently examined for posts containing dialect material, where the relevant post descriptions and comments were extracted manually. In particular, posts belonging to the “Mundart Monday” series were relevant because their descriptions contain Saxon text in Erzgebirgisch together with a directly translated German version. This constitutes the parallel data of the dataset.

3.3 Sachsen-Lese

The **major portion** of the monolingual Saxon material was collected from *Sachsen-Lese*, particularly from its sections containing dialect texts and songs, i.e., “Mundartliches” [10] and the songs of Anton Günther [11].

All HTML pages belonging to the selected sections were scraped, preserving the complete HTML of each webpage as the input for subsequent **LLM-based information extraction**: For each webpage, the complete HTML was provided to the model together with a structured extraction prompt [12]. The prompt instructed the model to identify the Saxon text and extract associated metadata, specified a certain format and included an example.

Two different models were used for this extraction task: *Llama-3.3-70B-Instruct* for the “Mundartliches” page and later *GLM-5.2-FP8* for the songs of Anton Günther. Both models were given the same extraction prompt. The Llama-based extraction generally recovered the relevant text, but some pages contained missing or incorrectly ordered passages, as well as incorrect publication years and source or origin information. The GLM-based extraction was more reliable for the pages to which it was applied, particularly for the text and author information, while year and origin information was also mostly correct. It needs to be noted, that the pages given to the GLM model were structured mostly uniform, while those given to the Llama model differ greatly.

The resulting textual output was subsequently parsed into JSON files following the intended dataset structure. These JSON files were then manually revised with a focus on correcting extraction errors such as omitted text, incorrect paragraph order, and erroneous metadata.

3.4 Additional Web Sources

To expand the collection beyond the main sources, additional webpages were identified through **targeted Google searches**. The searches used keywords including “sächsisch”, “sächsischer Dialekt”, “sächsisch Mundart”, “Sprachen in Sachsen”, “Mundart Sachsen”, “Sachsen Dialekt”, “sächsisch Übersetzungen”, and “Texte sächsisch”. For each query, the first ten result pages were examined for potentially relevant Saxon-language material.

These searches primarily yielded webpages containing relatively short passages or individual paragraphs of Saxon text. Relevant material was manually extracted and added to the dataset. As with the other non-Instagram sources, these webpages contributed only monolingual data.

3.5 OCR Experiments

To further expand the amount of historically relevant Saxon material, **scanned books** from the *Saxon State and University Library Dresden* (SLUB) [13] were investigated as a potential additional source. The SLUB catalogue was searched for suitable publications, with the aim of obtaining scanned and already transcribed books, but the existing full-text transcriptions associated with all relevant scanned books were found to contain substantial recognition artefacts. For example, the historical *s* (*f*) was recognized as *f*, *ü* as *ii*, and *Ä* as *L*. Because these substitutions were not sufficiently consistent within and across books, simple heuristic correction rules were not considered reliable enough to produce clean transcripts. **OCR** was therefore investigated as an alternative means of generating new transcriptions directly from the page images.

The OCR experiments used the scanned book pages as image input and tested two **OCR engines**, *Tesseract* [14] and *Kraken* [15]. The OCR pipeline rendered individual PDF pages at 400 dpi, converted them to grayscale, upscaled them, applied noise reduction, and deskewed the resulting images. For Tesseract, additional local contrast enhancement was applied. The recognized text was subsequently subject to basic post-processing, including whitespace normalization and correction of selected recurring Fraktur recognition errors. For Tesseract, the German (*deu*) and Fraktur (*frk*) language models were tested. Kraken was evaluated using the *german_print* and *austriannewspapers* recognition models. The same scanned pages were used as input, with the preprocessing differing slightly between the two OCR engines.

The resulting transcriptions were evaluated manually by comparing the OCR output with the original scanned pages. Both systems produced substantial **recognition errors**. Typical errors included interpreting image noise as characters or symbols, incorrectly assigning text to lines when spacing was ambiguous, and confusing visually similar characters, such as *M* and *A*, *H* and *MI*, or *ä* and *i*. Tesseract produced inaccurate recognition with both *deu* and *frk*, while Kraken produced fewer visible artefacts but remained similarly inaccurate overall with both models.

Consequently, none of the OCR-derived material was incorporated into the final dataset. The OCR experiments nevertheless established that the available scanned material could not be integrated reliably without substantially more extensive transcription or correction.

3.6 Dataset Representation and Metadata

The collected material was consolidated into a common dataset representation [16]. Each entry records the **Saxon text** and, where available, a **German translation**. Additional metadata fields record the **webpage** from which the entry was obtained, the **date of extraction**, the **source** (e.g. Instagram account, book title, author), the **regional origin** as specific as possible, the original **publication year**, and a **content tag**. The **length** of the Saxon text was also calculated and stored as an additional quantitative field. Source, origin and publication year often could not be determined exactly or at all. Entries without a tag are those texts that could not be assigned to a specific category, often small sections found on specific webpages.

The resulting dataset currently contains 200 entries with a total of 198,950 characters of Saxon text, of which 7 entries or 2,545 characters contain parallel data (see *Table 1*). Thus, the dataset is substantially larger as a collection of monolingual Saxon material than as a German-Saxon parallel corpus. *Table 2* shows that the dataset is rooted in at least 49 unique sources and 13 unique origins, although these origins belong partly to the same linguistic regions as established in Section 2.1. The collected material consists mostly of old poetry and literature. 80.5% of entries are poems and other older dialect texts, while more modern Saxon material is concentrated in Instagram posts (12.5%) and Wikipedia entries (4.5%) (see *Table 3*). *Table 4* showcases that at least 24% of the entries contain language from before 1990, but this number is supposedly significantly higher since most of the entries with a missing publication date are part of this historical poetry.

	Total	Parallel
Entries	200	7
Characters	198,950	2,545

Table 1: Overview

	Unique	Missing
Sources	49	27
Origins	13	81

Table 2: Source and Origin

Tag	Count	Share
poetry	161	80.5%
Instagram	25	12.5%
Wikipedia	9	4.5%
none	5	2.5%

Table 3: Tag distribution

Years	Entries	Share
≤ 1930	25	12.5%
≤ 1950	48	24.0%
≤ 1990	48	24.0%
≤ 2020	50	25.0%
> 2020	34	17.0%
Missing	116	58.0%

Table 4: Temporal distribution

4. Discussion and Conclusion

Applying LLM-based machine translation to Saxon faces several challenges beyond the general low-resource setting. Publicly available Saxon text is limited, and suitable German-Saxon parallel data is particularly scarce. This is problematic for MT because parallel data is required to directly learn correspondences between the two languages or varieties, while the low-resource setting already limits the amount of data available for model adaptation. In addition, there is not one uniform form of “Saxon”, but several regional varieties with differences in vocabulary and spelling. It can be assumed that this regional variation, together with the lack of standardized orthography, makes it more difficult for a model to learn consistent mappings and to generalize across regions.

The composition of the dataset introduces further limitations. Most of the collected material consists of older poetry and dialect literature, which does not necessarily reflect contemporary or colloquial Saxon. Vocabulary, grammatical patterns, and spelling may therefore differ from the language encountered in modern everyday communication, limiting the dataset’s suitability for MT towards modern Saxon. The parallel subset is particularly limited: only seven entries contain parallel data, all originating from Erzgebirgisch. This makes the parallel corpus both very small and regionally narrow. A model trained on such data may mainly learn patterns specific to Erzgebirgisch rather than mappings that generalize across the broader range of Saxon varieties. The parallel data also comes from a single source, which further increases the risk that source-specific language and translation patterns influence the resulting model.

While the LLM-based information extraction pipeline from Section 3.3 reliably distinguished Saxon from German text, it cannot be said whether this accounts for an existing capability of the used LLMs to detect Saxon, or if this distinction was based on the HTML structure.

Overall, the dataset is therefore more useful as an initial collection of Saxon text than as a representative training corpus for general German-Saxon MT. Further work would need to expand both the amount and regional diversity of contemporary parallel data.

5. Outlook

Several directions could further improve the dataset and enable actual MT experiments. A systematic review of existing public corpora was not feasible within the time available, but could reveal additional sources of Saxon text and potentially also parallel material. Proper OCR of historical publications could further increase the amount of monolingual data substantially, as the OCR experiments in this project showed that the available material would require more suitable recognition and correction before it could be integrated.

Social media is another promising source for contemporary Saxon, as it is more likely to contain informal and recent language than the historical literature dominating the current dataset. However, collecting such data at scale is difficult because of scraping restrictions and access limitations imposed by the platforms.

For future MT experiments, model adaptation and backtranslation could be used to make better use of the limited parallel data, as suggested by previous low-resource MT work. Another useful direction would be to test how well current LLMs already understand and

translate to Saxon without additional training. Evaluating this reliably would nevertheless be difficult without native-speaker judgements and/or a sufficiently large parallel dataset.

Declaration on Use of Generative AI

- For the LLM-based information extraction, the models were accessed via an API Key provided by *ScaDS AI*
- The *GLM-5.3-fast* model was used through the Chat Web Interface of *ScaDS.AI* as well as *ChatGPT* based on *GPT-5.5* for assistance in formulation of this report

Bibliography

- [1] Dialekte in Sachsen; Accessed: 21/08/2026; <https://dsa.info/sachsen/uebersicht>
- [2] Interview mit Dr. Evelyn Koch - So geht sächsisch.; Accessed: 21/08/2026; <https://www.so-geht-saechsisch.de/dialekt/interview-mit-dr-evelyn-koch>
- [3] Artemova, Plank; Low-resource Bilingual Dialect Lexicon Induction with Large Language Models; 2023
- [4] Lin et al.; Low-Resource Cross-Lingual Adaptive Training for Nigerian Pidgin; 2023
- [5] de Gibert et al.; Scaling Low-Resource MT via Synthetic Data Generation with LLMs; 2025
- [6] Pei et al.; Understanding In-Context Machine Translation for Low-Resource Languages: A Case Study on Manchu; 2025
- [7] Su et al.; Unlocking Parameter-Efficient Fine-Tuning for Low-Resource Language Translation; 2024
- [8] Saxon Wikipedia; Accessed: 23/05/2026; https://web.archive.org/web/20240808042125/https://incubator.wikimedia.org/wiki/Wp/sxu/Main_Page
- [9] @simplysaxony Instagram Account; Accessed: 14/06/2026; <https://www.instagram.com/simplysaxony/>
- [10] Mundartliches; Accessed: 29/05/2026; https://www.sachsenlese.de/index.php?article_id=89
- [11] Lieder von Anton Günther; Accessed: 20/07/2026; https://www.sachsenlese.de/index.php?article_id=123
- [12] Prompt; Repository: https://github.com/Knowledge-aware-AI/saxon-corpus/blob/main/src/llm/prompts/extract_from_html.md
- [13] SLUB Dresden; <https://www.slub-dresden.de/>
- [14] Tesseract OCR; Accessed: 21/07/2026; <https://tesseractocr.org/>
- [15] kraken; Accessed: 21/07/2026; <https://kraken.re/main/index.html>

- [16] Dataset; Repository: <https://github.com/Knowledge-aware-AI/saxon-corpus/blob/main/data.csv>